

Short Answer Questions

Unit 1: Computer System

1. Define the Computer.
2. How do IPOS work?
3. What is GIGO?
4. List any four computer features.
5. Why is a computer called a diligence machine?
6. List four computer-related uses.
7. Identify distinct computer storage units.
8. What is the primary function of the CPU in a computer system?
9. Explain the difference between RAM and ROM in terms of functionality.
10. How does a hard drive differ from RAM in terms of data storage?
11. What is the role of a motherboard in a computer system?
12. Explain the concept of cache memory and its importance in CPU performance.
13. Define the term “bus” in the context of computer architecture.
14. What is the purpose of an input device in a computer system? Provide examples.
15. What is a peripheral device, and how does it enhance a computer system?
16. What does the term “open source” mean in software?
17. How does open-source software differ from proprietary software?
18. What is system software, and what role does it play in a computer system?
19. Define application software and provide examples of common applications.
20. What is the purpose of utility software, and give examples of utilities?
21. What are the types of computers based on working principles?
22. Write any two differences between impact and non-impact printers. (Model Set 1)
23. Why is a computer called an information processing machine?
24. Differentiate between licensed and open source software.
25. Describe the roles of Control Unit and Register.
26. What are ports in a computer system? Explain with any two examples.
27. Why are registers considered faster storage or execution than RAM?
28. Explain why primary memory is called volatile.
29. Explain why the motherboard is often called the “backbone” of a computer.
30. What is the difference between a monitor and a printer?
31. Differentiate between primary memory and secondary memory. (Model Set 6)
32. Why is software essential for computer functionality?
33. What is optical storage? Write with examples.
34. What is a printer? What are its types?

35. What is meant by versatility in terms of features of a computer?
36. Why is the QWERTY keyboard layout designed in its current form?
37. Explain the functions of the data bus in a computer system.
38. Why is the SSD more popular than the hard disk in a modern computer today?
39. Which output device would be most suitable for displaying a digital presentation? Explain why.
40. How does system software differ from application software? (Model Set 10)
41. What is the purpose of utility software? Give examples of Utility software (any two).
42. Write down the difference between impact and non-impact printer (any two). (Model Set 11)
43. Write any two differences between Primary Memory and Secondary Memory.
44. You are watching an educational video in the computer lab but cannot hear any sound. Which output device should you check, and why?
45. Write the functions of ALU and CU.
46. Write any two differences between SRAM and DRAM.
47. Why is "diligence" considered an important characteristic of a computer?
48. Suggest two real-life situations where computers improve efficiency.
49. List any four characteristics of a computer and explain any one.
50. Suppose you are designing a multimedia classroom. Which two output devices would you choose and why?
51. Explain the difference between RAM and ROM. (Asmita Set 4)
52. Recommend one scenario where tailored software is more useful than packaged software.
53. Why is primary memory considered volatile? (Asmita Set 5)
54. Explain how open-source software can be applied in educational institutions.
55. Differentiate between system software and application software. (Asmita Set 6)
56. A bank wants to ensure secure customer records. Which type of memory (primary/secondary) is more suitable for long-term storage and why?
57. What is open-source software? State any two advantages.
58. You are advising a graphic designer. Which input and output devices would you recommend and why?
59. Explain the role of the Control Unit in a computer system.
60. An airline company needs to process huge amounts of booking data. Which type of software would you recommend (system/application/tailored) and why?
61. Describe the importance of accuracy as a characteristic of a computer.
62. If the RAM of a computer is too small, what problems might a user face in daily activities?
63. Why is versatility considered an important feature of computers? (Asmita Set 10)
64. You are setting up a small office. Which type of printer (inkjet, laser, or 3D) would you recommend and why?
65. Explain the difference between primary memory and secondary memory with examples.
66. A student claims that USB drives are primary memory. How would you correct this misunderstanding with an explanation?

Unit 2: Number System

1. Decimal to Binary: $(56)_{10}$, $(78)_{10}$, $(123)_{10}$, $(345)_{10}$, $(540)_{10}$, $(572)_{10}$, $(546)_{10}$, $(1098)_{10}$, $(2103)_{10}$, $(445)_{10}$
2. Binary to Decimal: $(1101)_2$, $(1010)_2$, $(10010)_2$, $(10110)_2$, $(101001)_2$, $(11100111)_2$, $(111100)_2$, $(10010011)_2$, $(1011100)_2$, $(100110)_2$
3. Decimal to Octal: $(69)_{10}$, $(216)_{10}$, $(767)_{10}$, $(79)_{10}$, $(443)_{10}$, $(413)_{10}$, $(765)_{10}$, $(1334)_{10}$, $(1825)_{10}$, $(2783)_{10}$
4. Octal to Decimal: $(124)_8$, $(242)_8$, $(333)_8$, $(763)_8$, $(103)_8$, $(451)_8$, $(3401)_8$, $(1045)_8$, $(438)_8$, $(611)_8$
5. Decimal to Hexadecimal: $(55)_{10}$, $(540)_{10}$, $(225)_{10}$, $(880)_{10}$, $(2046)_{10}$, $(2024)_{10}$, $(6678)_{10}$
6. Hexadecimal to Decimal: $(56)_{16}$, $(67)_{16}$, $(558)_{16}$, $(B74)_{16}$, $(20D3)_{16}$, $(DEF)_{16}$, $(6E3)_{16}$, $(63F)_{16}$
7. Binary to Hexadecimal: $(1000110)_2$, $(11001)_2$, $(1111000)_2$, $(11110000111)_2$, $(101010110)_2$, $(1110010110)_2$, $(11011001)_2$, $(1001100)_2$
8. Hexadecimal to Binary: $(D4)_{16}$, $(643)_{16}$, $(189)_{16}$, $(2BF)_{16}$, $(A9F)_{16}$, $(FACE)_{16}$, $(FB4)_{16}$, $(1B2)_{16}$
9. Convert $(10011)_2$ into decimal number system and $(105)_{10}$ into octal number system; Perform $(1011-101)_2$ and $(1101)_2 + (1100)_2$ binary operations. (Model Set 1)
10. Convert $(1101101010)_2$ into decimal and $(1562)_{10}$ into octal; Perform $(111010+10100)_2$ and $(110011-10100)_2$. (Model Set 2)
11. Convert $(7533)_8$ into decimal and $(2989)_{10}$ into binary; Perform $(11011 \times 1011)_2$ and $(111010 \div 110)_2$. (Model Set 3)
12. Convert $(1001101110)_2$ into hexadecimal and $(12BF)_{16}$ into binary; Perform $(1001011+110010)_2$ and $(101101 \times 11001)_2$. (Model Set 4)
13. Convert $(CADA1)_{16}$ into decimal and $(10101100101110)_2$ into octal; Perform $(10100-1011)_2$ and $(111111 \times 111)_2$. (Model Set 5)
14. Convert $(1011101)_2$ into octal and $(5121)_8$ into binary; Perform $(1010011 \times 11011)_2$ and $(1000110 \div 111)_2$. (Model Set 6)
15. Convert $(6451)_8$ into hexadecimal and $(A90F)_{16}$ into octal; Perform $(1010010-11011)_2$ and $(1000101 \times 1101)_2$. (Model Set 7)
16. Convert $(1110010)_2$ into hexadecimal and $(12345)_{10}$ into octal; Perform $(1001011 \times 1101)_2$ and $(10110011 \div 110)_2$. (Model Set 8)
17. Convert $(BADE)_{16}$ into octal and $(23045)_{10}$ into hexadecimal; Perform $(100111+11011)_2$ and $(1010011 \times 1001)_2$. (Model Set 9)
18. Convert $(71373)_8$ into binary and $(1011100111)_2$ into hexadecimal; Perform $(1001101 \times 11001)_2$ and $(10010011 \div 111)_2$. (Model Set 10)
19. Convert $(1101)_2$ to $(?)_{10}$ and $(7601)_8$ to $(?)_{16}$; Perform $(110111)_2 + (11001)_2$ and $(10101)_2 - (11001)_2$. (Model Set 11)
20. Convert the decimal number $(45)_{10}$ into binary and $(426)_8$ into decimal; Perform $(1110)_2 + (1011)_2$ and $(1001)_2 - (110)_2$. (Model Set 12)
21. Convert $(765)_{10}$ to octal and $(AFE)_{16}$ to binary; Perform $(101001)_2 + (1111)_2$ and $(111101)_2 - (1001)_2$. (Model Set 13)
22. What is a number system?
23. Convert $(324)_{10}$ into binary.

24. Define the binary number system with an example.
25. Convert $(1011)_2$ into the decimal number system.
26. What is the octal number system? (Asmita Set 4)
27. Convert $(1357)_{10}$ into octal.
28. State the base of the hexadecimal number system.
29. Convert $(217)_8$ into the decimal number system.
30. What is the decimal number system? (Asmita Set 6)
31. Convert $(876)_{10}$ into hexadecimal.
32. List any two applications of the number system in daily life.
33. Convert $(CAD)_{16}$ into the decimal number system.
34. Differentiate between the binary and decimal number systems. (Asmita Set 8)
35. Convert $(11111011)_2$ into octal.
36. Differentiate between octal and hexadecimal number systems.
37. Convert $(3547)_8$ into hexadecimal number system.
38. Explain why computers use the binary number system.
39. Convert $(7BCD)_{16}$ into the octal number system.
40. Why is the binary number system important for computers?
41. Perform binary division: $(100001)_2 \div (101)_2$.

Unit 3: Block Programming

1. What is scratch?
2. List down the components of Scratch interface.
3. What is a micro:bit? Give any two applications of micro:bit.
4. How does a micro:bit help in physical computing?
5. What is ATmega328P?

Unit 4: Web Technology

1. What is DNS?
2. Define HTML tag.
3. What is the main function of the <HEAD> tag?
4. What is an attribute?
5. Which tag is used for breaking lines or paragraphs in HTML?
6. What is an Empty tag?
7. What is a container tag?
8. Write down the syntax of the comment tag.
9. What is the output of $(a+b) <\sup>2</sup>$ tag?
10. What is the function of the <TR> tag?

11. Write down the use of Form in web pages.
12. Create a table having two rows and two columns using HTML tags. (Model Set 1)
13. Write HTML code to display one text input and a reset button using HTML. (Model Set 2)
14. Create an ordered list of four favorite subjects using HTML. (Model Set 3)
15. How do you display an image using HTML? (Model Set 4)
16. Write HTML code to display a short paragraph about "My Hobby". (Model Set 5)
17. Explain the internal style sheet and the external style sheet. (Model Set 6)
18. Write HTML code to make a horizontal line on the webpage. (Model Set 7)
19. Name three ways to apply CSS to an HTML page. (Model Set 8)
20. Explain the basic tags of HTML. (Model Set 9)
21. Write HTML code to create a hyperlink to www.google.com. (Model Set 10)
22. Write HTML code to create a heading of "Welcome to My Blog". (Model Set 12)
23. Write down the general structure of an HTML document. (Model Set 13)
24. Write the steps to create, save, and view an HTML file.
25. Write HTML code to display "Welcome to Web Technology" in bold and underlined text.
26. Create an ordered list of five fruits in HTML.
27. Create an unordered list of five programming languages.
28. Write HTML code to insert an image named "logo.png" with width 200px and height 100px.
29. Write the HTML code to create a hyperlink to www.openai.com with the text "Visit OpenAI".
30. Write HTML code to create a table with 2 rows and 3 columns.
31. Write CSS code to change the background color of a webpage to light blue.
32. Write CSS code to set the font size of all paragraphs to 16px.
33. Write HTML and CSS code to display a red heading of size 24px.

Unit 5: Internet and Social Media

1. What is the Internet? Write its two uses.
2. What is a WWW? List out the components of www.
3. What is a web browser? Give an example.
4. Write down some examples of online meeting platforms.
5. What are Web Protocols? List out the examples of the Web Protocols?
6. Write down some uses of email?
7. Define social media. List any two examples of social media platforms. (Model Set 1)
8. What are two major benefits of using blog in education? (Model Set 1)
9. What is the internet? Write one use. (Model Set 2)
10. Write two features of blog. (Model Set 2)
11. What is a web browser? Give examples. (Model Set 3)
12. Define protocol with examples. (Model Set 3)

13. Write two uses of email. (Model Set 4)
14. Define social media with examples. (Model Set 4)
15. Define blog. Write two advantages of blog. (Model Set 5)
16. Why is HTTPS more secure than HTTP? Give one reason. (Model Set 5)
17. Define Blog. Write down any two disadvantages of blog. (Model Set 6)
18. Write two differences between HTTP and HTTPS. (Model Set 6)
19. Define Social media. List two uses of social media. (Model Set 7)
20. Explain remote login with examples. (Model Set 7)
21. How can we use social media in a safe way? (Model Set 8)
22. Define the concept of WWW. (Model Set 8)
23. Write any two advantages of remote login. (Model Set 9)
24. Write two advantages of social media. (Model Set 9)
25. Write two disadvantages of social media. (Model Set 10)
26. List the components of a URL. (Model Set 10)
27. Define web browser. Give any two examples. (Model Set 11)
28. Write down the advantages and challenges of social media (two/two each).
29. "Internet is a powerful learning tool, but it can also be harmful". Justify the statement with two appropriate points.
30. Write any two positive impacts of social media on a student's learning.
31. Define the term Browser and URL.
32. Define WWW with one example of its use.
33. How does the Google Search Engine work in retrieving information?
34. Write two differences between a browser and a search engine.
35. Differentiate between HTTP and HTTPS, highlighting the advantages of each.
36. Explain the term URL with an example.
37. You are working from home and need to access important files stored on your school computer. How would you use remote login to accomplish this task? Explain the steps briefly.
38. Write two advantages of Remote login.
39. What is HTTPS? Why is it considered more secure than HTTP? (Asmita Set 6)
40. Compare AnyDesk and TeamViewer in terms of their uses.
41. List any two features of Google Meet.
42. How would you join and host a meeting on Zoom?
43. What are protocols? Give two examples.
44. Explain with steps how to create a free blog on WordPress or Blogger.
45. Write two advantages of using email in business communication.
46. What is a blog? List two features of a blog. (Asmita Set 10)
47. Your school is planning to conduct online classes and meetings. As an IT student, which platform would you recommend between Zoom and MS Teams, and why? Support your answer with one feature of each.

48. Define video conferencing and write one of its benefits.
49. How can blogs be effectively used for educational purposes?

Unit 6: Cyber Security and Digital Citizenship

1. What is Cybersecurity?
2. Explain Cyber Law by relating it with the constitution.
3. What is a Firewall? Why is it different from antivirus software?
4. You use digital devices on a daily basis. What do you know about digital society?
5. What is malware? Write down the different types of malwares.
6. You want to buy a new headset from an unfamiliar online store that is offering a 90% discount, but the website has no contact address or customer reviews. What two risks are involved in shopping from such a site?
7. Write any two preventive measures of cyber crime.
8. Write two safe practices for accessing websites securely.
9. Discuss two ways social media can be misused and suggest solutions.

Unit 7: Concept of Programming

1. A set of instructions to perform a specific task.
2. The set of rules to write a program.
3. Language translator that reads and translates program code to machine code line by line.
4. The graphical or pictorial representation of a program.
5. Text added in source code for the purpose of providing information or explanation that is ignored by the Python interpreter during execution.
6. A data type that only provides true or false value.
7. A special symbol that is used to check the relationship between two variables.
8. The values or entities that operators act upon in programming.
9. A decision-making control structure that helps our program to choose what to do.
10. The process of repeating a task until a specified condition is met.
11. Programming language is used to give commands to the computer. (True/False)
12. Computers understand high level programming easily. (True/False)
13. Compiler and interpreter translate programming code to machine code. (True/False)
14. Algorithms are a graphical representation of instructions. (True/False)
15. Python is a machine understandable programming language. (True/False)
16. Print statement is used to display output in Python. (True/False)
17. In Python, "int" data type denotes alphabetical characters such as hello. (True/False)
18. Type casting is the conversion of one data type to another. (True/False)
19. In AND operator all the conditions must be true for the result to be true. (True/False)

- 20.** In Python, if statement is used to provide a condition that, when false, allows a block of code to be executed. (True/False)
- 21.** A for loop in Python is used to repeat a process until a specified time given by the user. (True/False)
- 22.** Dictionaries have key : value pairs. (True/False)
- 23.** List items cannot be changed. (True/False)
- 24.** In the dictionary the data can be deleted by value. (True/False)
- 25.** Study the Python code (a=2; for k in range(1,8): print(a); a//=10). (a) How many times does the loop execute? (b) List the integer type variables. (Model Set 1)
- 26.** Write the output of the given program and show it in a dry run table (import math; n=13; d=0; p=0; while n!=0: c=n%2; d=d+c*pow(10,p); p+=1; n//=2; print the final output). (Model Set 1)
- 27.** Write a Python program to enter any three numbers and display their average. (Model Set 1)
- 28.** Study the Python code (a=25; b=4; c=a%b; print(c)). (a) What will be printed? (b) Which operator is used to find the remainder? (Model Set 2)
- 29.** Write the output of the given program and show it in a dry run table (s=0; for n in (2,1,6,4,5,7): if n%2==0: print(n); s=s+n; print("sum",s)). (Model Set 2)
- 30.** Study the Python code (val=int(input("Enter a value")); if val>0: print("Positive"); else: print("Negative or Zero")). (a) What will happen if the user enters -1? (b) What is the data type of val? (Model Set 3)
- 31.** Study the Python code (word="HELLO"; print(word.lower())). (a) What is the output? (b) Which string function is used here? (Model Set 4)
- 32.** Study the Python code (subject=["computer","maths","science"]; for L in subject: print(L)). (a) How many items are in the list? (b) What will be the second output? (Model Set 5)
- 33.** Study the Python code (for i in range(2,11,2): print(i,end=" ")). (a) Which numbers will be printed? (b) How many times does the loop execute? (Model Set 6)
- 34.** Rewrite the given program after correcting the bug (n=10; for m range(1,10): print(n,end="*"); n=n+1). (Model Set 7)
- 35.** Rewrite the given program after correcting the bugs (import math; n=str(input("Enter a number")); s=math.sqrt(s); print "Square root of",n,"=",n). (Model Set 8)
- 36.** Rewrite the given program after correcting the bugs (n="nepal"; print(upper(n)); print(lower(n)); print(n.len())). (Model Set 9)
- 37.** Re-write the given program after correcting the bugs (for i range[1,6]: print(i); if i%2==0: print("Even")). (Model Set 10)
- 38.** Write down the output of the given piece of Python code with a dry run table (for l in range(1,10): if(l%2==0): print(l**2,end=" ")). (Model Set 11)
- 39.** Study the Python program and answer: (n=1234; sum=0; while n!=0: R=n%10; sum=sum+R; n=n//10; print("Sum of digits=",sum)). (a) How many times the loop body will execute? (b) List out all the numeric variables used in above program. (Model Set 11)
- 40.** Study the following Python code and answer the questions below (count=1; while count<=5: print(count); count+=2). (a) How many times does the loop run? (b) List all the variables used in above program. (Model Set 12)
- 41.** Study the following Python code and answer (celsius=float(input("Enter the temperature in celsius:")); fahrenheit=(celsius*9/5)+32; print("Temperature in Fahrenheit:",fahrenheit)). (i) What will be the output if the value of celsius is 36? (ii) List different data types used in the above program. (Model Set 13)

42. Write the output of the given program and show them in a dry run table (for num in range(1,5): print(num)). (Model Set 13)
43. Write a python program to check if a number is even or odd. (Model Set 13)
44. What are the types of programming languages?
45. Write a Python program to calculate the simple interest.
46. Differentiate between a compiler and an interpreter with examples.
47. List two advantages of third-generation programming languages (3GL).
48. Write a Python program to check whether a number is positive, negative, or zero. (Asmita Set 3)
49. Write the differences between the 'for' loop and the 'while' loop.
50. Define algorithm.
51. Write a program to input a number and check if it is even or odd.
52. Differentiate between int, float, and string with examples.
53. Define flowchart and list two flowchart symbols.
54. Write a Python program to calculate the sum of two numbers entered by the user.
55. Write any two advantages of Python.
56. What is Python? Write two features of it.
57. Write a program to display numbers from 1 to 100 using a loop.
58. Compare assembler and interpreter.
59. What is a variable in Python?
60. Write a Python program to find the greatest of three numbers.
61. Why is debugging important in programming? Explain briefly.
62. What is a compiler?
63. Write a program to calculate the area of a rectangle.
64. Explain the use of library functions in Python with examples.
65. What is type casting?
66. Write a Python program to convert Celsius to Fahrenheit.
67. What will be the output of the Python program: x=5, y="5", print(x + int(y))?
68. Name two conditional statements in Python.
69. Write a Python program to display all even numbers up to 50.
70. Explain the concept of string formatting in Python.
71. List three data types in Python.
72. Write a Python program to calculate the factorial of a number.
73. Discuss the importance of variables in programming with an example.